

Function Spaces
B. Math (Hons.) III Year
Backpaper Exam, December 2025

Maximum Score: 50

Duration: 3 hours

1. State Arzela-Ascoli Theorem and use it to prove the following:

Let $f_n : [0, 1] \rightarrow \mathbb{R}$ be a sequence of continuous functions such that

$$|f_n(x)| \leq 1 \quad \forall x \in [0, 1] \text{ and } \forall n \in \mathbb{N}.$$

Define a sequence of functions $F_n : [0, 1] \rightarrow \mathbb{R}$ by

$$F_n(x) = \int_0^x f_n(t) dt \quad \forall n \in \mathbb{N} \quad (x \in [0, 1]).$$

Show that $\{F_n\}$ has a subsequence that converges uniformly on $[0, 1]$. [Marks: 6]

2. Let $f : [0, 1] \rightarrow \mathbb{R}$ be Riemann integrable and consider the equation

$$f(x) = x + \lambda \int_0^1 (x^2 + y^2) f(y) dy \quad (\lambda \in \mathbb{R}).$$

Show that, the above equation has a unique solution in $C[0, 1]$ if $|\lambda| < \frac{1}{2}$. [Marks: 6]

3. State Stone-Weierstrass Theorem and use it to prove that every continuous function $f \in C[0, 1]$ can be uniformly approximated by a sequence of even polynomials. [Marks: 6]

4. Consider $\mathcal{M} = \{E \subseteq \mathbb{R} : E \text{ is countable or } E^c \text{ is countable}\}$. Show that,

(a) $\mathcal{M}$ is a σ -algebra.

(b) $\mathcal{M} \subsetneq \mathcal{B}_{\mathbb{R}}$, where $\mathcal{B}_{\mathbb{R}}$ is the Borel σ -algebra on $\mathbb{R}$. [Marks: 6]

5. (i) State Monotone Convergence Theorem.

(ii) Show by an example that Monotone Convergence Theorem may not hold for decreasing sequence of functions. [Marks: 6]

6. Consider the 2π -periodic function f on $\mathbb{R}$ such that

$$f(x) = \begin{cases} -x, & -\pi \leq x < 0 \\ x, & 0 \leq x < \pi. \end{cases}$$

Find the Fourier series of f and using it deduce that

$$1 + \frac{1}{3^2} + \frac{1}{5^2} + \cdots = \frac{\pi^2}{8}$$

[Marks: 8]

7. State Fejer's Theorem on Fourier series convergence and use it to conclude that continuous 2π -periodic function on $\mathbb{R}$ can be uniformly approximated by the sequence of trigonometric polynomials. [Marks: 6]

8. Considering the function $f(x) = x$ on $[-\pi, \pi)$ and using the Parseval's identity deduce that

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}.$$

[Marks: 6]
